



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Software Engineering

**AIUB Portal Redefined: Integrating Campus Life with
Smart Digital Innovation**

Submitted To

Tonny Shekha Kar

Project Planning Report

Submitted By

Semester: Summer 2024-2025		Section: D	Group Number: 01	
SN	Student Name	Student ID	Contribution (CO3)	Individual Marks
01	SADIA ISLAM	23-51943-2		
02	ABDULLAH AL TAMJID	23-51887-2		
03	ARNOB SARKER SUPTA	23-52080-2		
04	NILOY PAUL	23-51773-2		
05	KAHIUM AHAMED FAHIM	22-48593-3		

Project Estimation(COCOMO) :

The AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation project has adopted the Organic COCOMO (Constructive Cost Model) mode due to the well-defined project requirements and manageable system complexity.

This mode is deemed most suitable as it facilitates realistic estimations of effort and resources for projects characterized by structured objectives and relatively low innovation risk. The Organic COCOMO model is specifically tailored for projects where the development team is experienced, the project scope is clearly delineated, and the technical challenges are within the team's expertise, aligning with the nature of the AIUB Portal Redefined initiative.

The application of the Organic COCOMO model is also informed by the project's timeline and resource constraints, as evidenced by the test execution dates ranging from May 1, to September 30, 2025.

This short development window necessitates a model that provides reliable effort and schedule estimates, which the Organic COCOMO achieves by leveraging historical data from similar project.

For **Organic** type,

P : project complexity = 1.05, **T** : SLOC-dependent coefficient = 0.38,

SLOC : source lines of code = 2450 lines (2.45 KSLOC) , **Coefficient**_{<Effort Factor>} = 2.4

Effort = PM = Coefficient_{<Effort Factor>}***(SLOC/1000)^P**

$$= 2.4 * (2450 / 1000)^{1.05}$$

$$= 6.25 \text{ Person-Months}$$

Development time = DM = 2.50*(PM)^T

$$= 2.50 * (6.25)^{0.38}$$

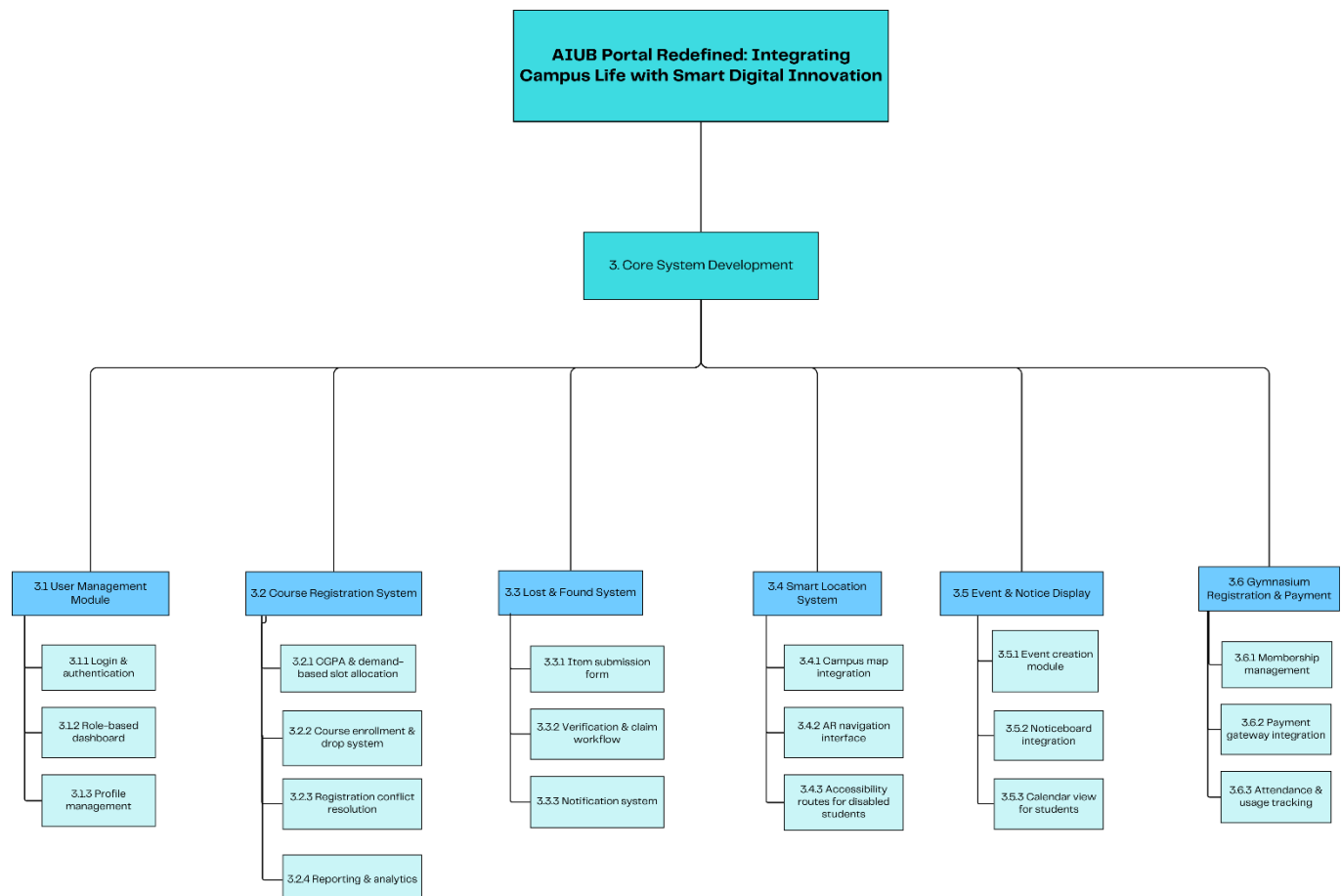
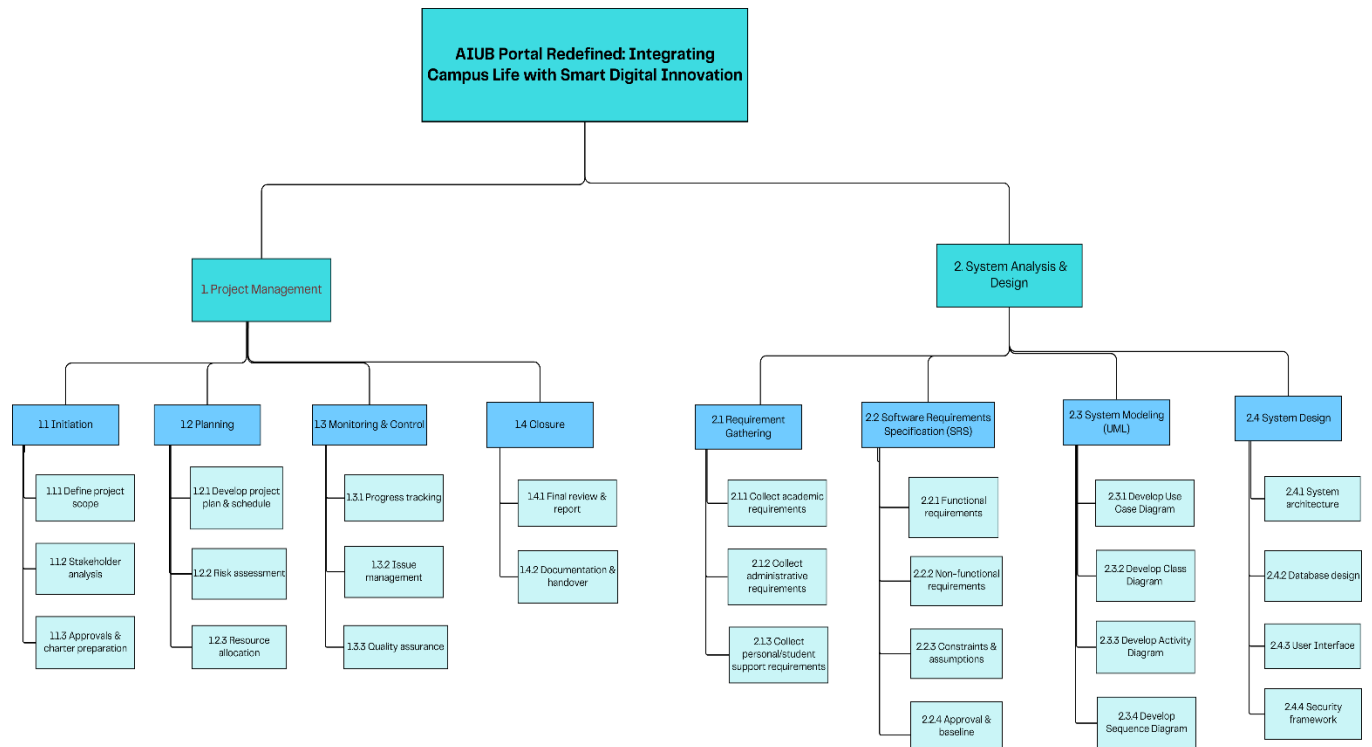
$$= 5.0 \text{ Months}$$

Required number of people = ST = PM/DM

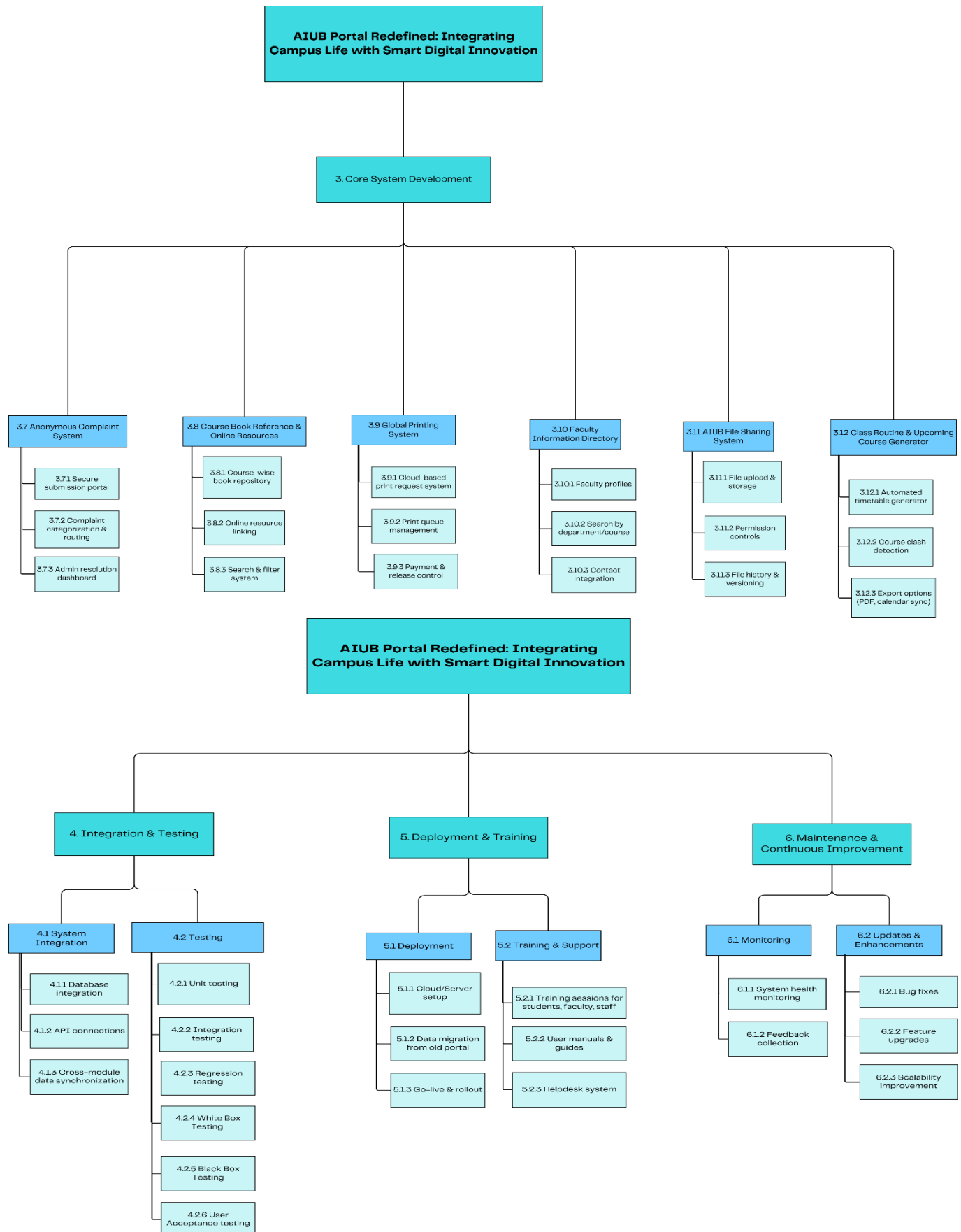
$$= 6.25 / 5.0$$

$$= 1.25 \approx 2 \text{ Persons}$$

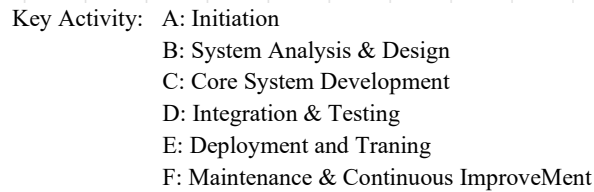
Work Breakdown Structure (WBS): A Work Breakdown Structure (WBS) is a hierarchical decomposition of a project into smaller, manageable tasks or deliverables. It helps organize and define the total scope of work, making planning, execution, and tracking easier.



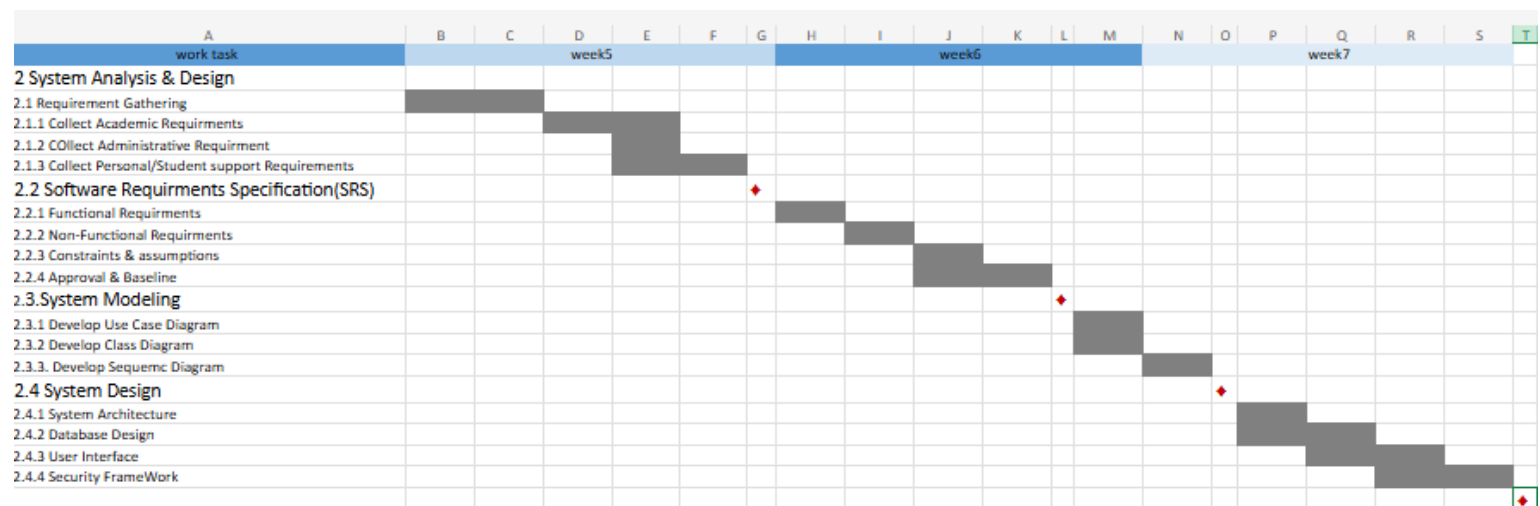
Work Breakdown Structure (WBS) Continued:



Overview Timeline:



Detailed Timeline :



Detailed Timeline Continued:

